National University of Singapore

Institute of Systems Science
Intelligent Reasoning Systems
Project report

GROUP PROJECT REPORT · SHORT VERSION

AnchorAI

Explainable, budget-aware retention planning for gyms

Churn prediction · uplift estimation · rule-based reasoning · case-based reasoning
genetic algorithm with an exact CP-SAT benchmark · knowledge graph · grounded staff assistant

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Abstract

About one gym member in three leaves each year, and most stop attending weeks before they cancel. Retention tools flag members at risk but leave the decision that matters (whom to contact this week, with which offer, within a fixed budget and limited staff hours) to guesswork. AnchorAI makes that decision and explains it. A calibrated ensemble predicts churn on real members; a forward-chaining rule base encodes the gym's playbooks and guardrails; case-based reasoning estimates what each offer would achieve for each member; a genetic algorithm, benchmarked against an exact CP-SAT solver, builds the weekly plan; a knowledge graph adds signals the churn model cannot see; and a grounded assistant answers staff questions from the system's own evidence. Segmentation and churn are evaluated on 4,000 real members of a public dataset; decisions on a documented synthetic scenario layer whose true offer effects are known, which shows that the method works when effects are of this kind, not how large they are in a real gym.

0.908
churn F1 on 800 unseen real members (PR-AUC 0.973)

+22–138%
more members kept than contacting by risk or by schedule, at equal budget

0.24%
largest gap of the genetic algorithm to the proven optimum

96.2%
of the assistant's judged sentences supported by its evidence

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1 Business case, market research and objectives

Gym members rarely cancel on impulse: they stop coming first and cancel weeks later, and that lag is the window in which a well-chosen contact can still keep them. The median member retention rate was 66.4% in 2024 across 175 operators [1], and new members are the most fragile: in one fitness centre 63% stopped before their third month [2]. Current practice produces a list or a score but not a decision. Calendar campaigns spend on members who would have stayed, threshold rules ignore trends, and vendor risk scores give staff a number they cannot interrogate. The proposal's market research found that no vendor publicly exposes constrained allocation, that vendors target risk although targeting by expected response reduces churn more [3], and that the intelligence layer is priced for chains of five or more clubs. AnchorAI is therefore positioned as an *explainable, budget-aware retention planner for*

operators with one to five clubs. No real operator was involved: the interviews that shaped the rules were simulated, and every comparison is against baseline policies in our own experiments.

Table 1 Objectives, how each is tested, and the verdicts argued in Section 6.

	Objective	Test and baseline	Verdict
O1	know who is at risk, and why	E1 segments (every k); E2 churn (visit rule, logistic regression); E7 graph signals	● Met
O2	know whom a contact would keep	E3 uplift (T-learner, risk, random)	● Partly met
O3	plan the week within budget, staff and stock	E4 plan value (contact-all, risk-top-k); E5 optimality and speed (CP-SAT)	● Met
O4	keep people in control	E6 rules and guardrails; tests of explanation and audit coverage	● Met
O5	usable by staff in plain language	E8 assistant (keyword router)	● Partly met
O6	reproducible and dependable	fresh rebuilds; tests with services down	● Met

2 System design

One web application serves administrators, managers and staff; one FastAPI service checks every request's role, audits every change and runs long jobs in the background; and three reasoning components do the work (Figure 1). They meet at three contracts: *data tables* written once by the data build and read by everything else; an *allowed-offer list* from the rule engine, so the optimiser never sees an offer the rules block; and an *explanation record* per plan item (drivers, rule trace, graph signals, ten similar past cases), which the plan's 'why' panel and the assistant both read, so they always give the same reason. Each week runs from scores to an approved, assigned call list, and logged outcomes feed the next week (Figure 2).

3 Data and the scenario layer

Segmentation and churn prediction use only the 4,000 real members of the public *Gym Customers Features and Churn* dataset [4]: 14 columns, 26.5% churners, split 3,200/800 with stratification, and four derived features such as the visit trend (this month's visits over the lifetime rate), whose median is 0.69 for members who left and 1.00 for those who stayed (Figure 3). Two same-month columns may leak the label, so E2 reports the model with and without them. The snapshot has no clubs, classes, offers or contact history, so a documented *scenario layer* attaches a synthetic gym: three clubs, nine instructors and 18 classes consistent with each member's real visits; four offers with costs, staff minutes and stock; a heterogeneous ground-truth uplift for all 16,000 member–offer pairs (mean +5.8 points); two gym events (an instructor leaving and a class cancelled); and a randomised history of 2,400 past contacts (Figure 4). The ground truth is used only to grade, never by an estimator or planner. One command rebuilds every table, model and score in about 20 seconds from one seed.

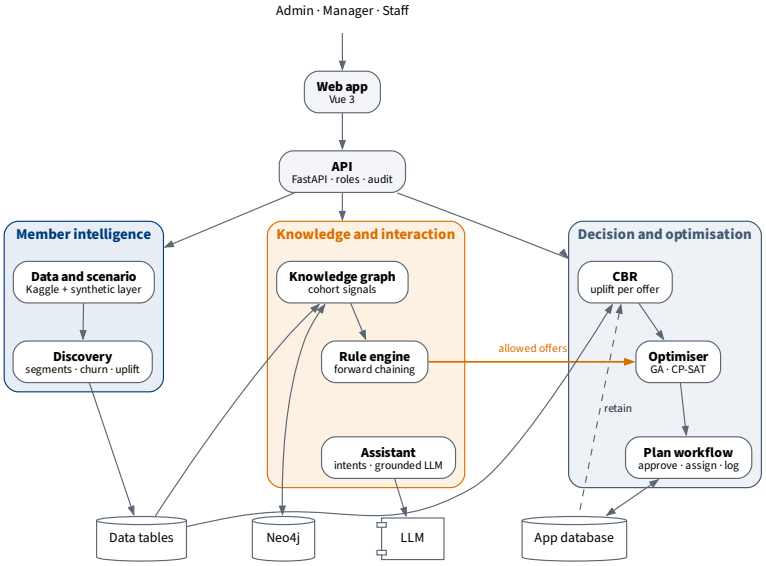


Figure 1 Architecture. Arrows show data flow; the dashed arrow is the learning loop through which logged outcomes return to the case base.

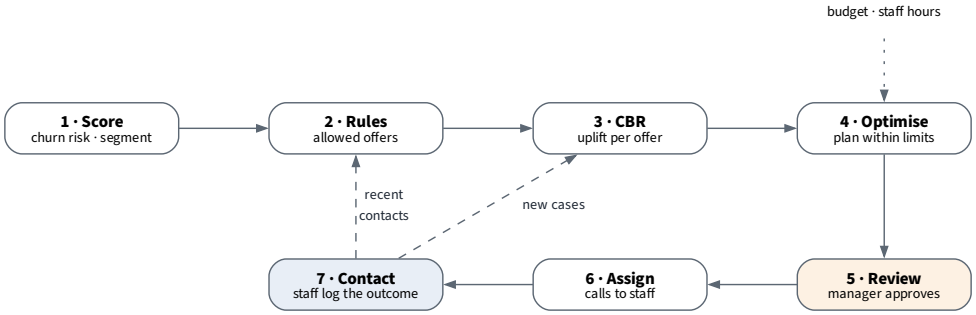


Figure 2 The weekly cycle: steps 1–4 run automatically within the week's budget and staff hours; the manager reviews and approves, and staff contact members and log outcomes.

4 Methods

4.1 Member intelligence

K-means [5] on seven behavioural features gives segments that staff can name; k is chosen by a fixed rule on the silhouette [6] and names come from rules on the centres, so they survive a rebuild. Churn is predicted by the mean of a class-weighted logistic regression and LightGBM [7] over 17 features (Figure 5). Class weights matched SMOTE-NC [8] in cross-validation and were kept as simpler; a Platt calibrator [9] fitted on a held-out slice makes the probabilities honest, which matters because the planner multiplies them by uplift and value. Training members are scored out of fold, and each

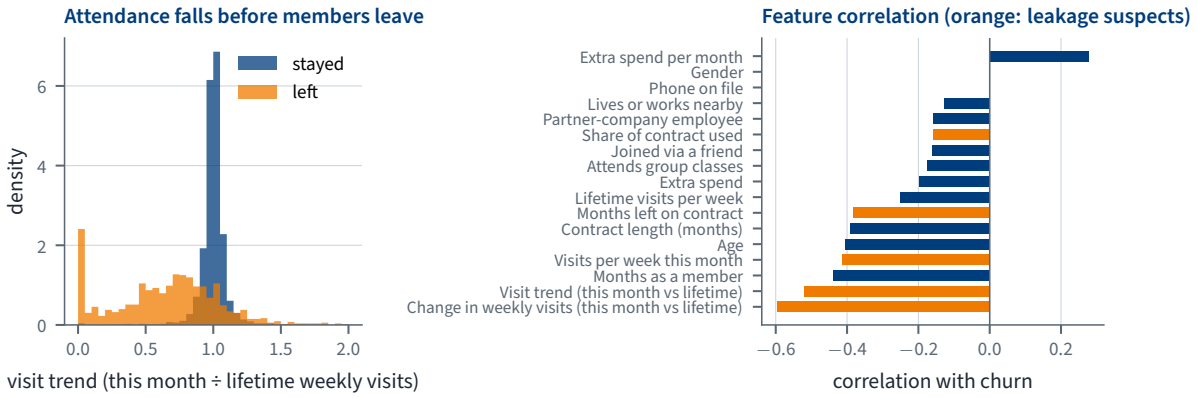


Figure 3 Left: the visit trend of members who stayed clusters at 1, while members who left spread below it, with a spike at 0 for those who did not come this month (medians 1.00 and 0.69). Right: correlation of each feature with churn; the two strongest are leakage suspects. Real Kaggle data.

Source: `scripts/figures.py` (`trend`), as in `a_features_leakage.ipynb`, cells 2–4.

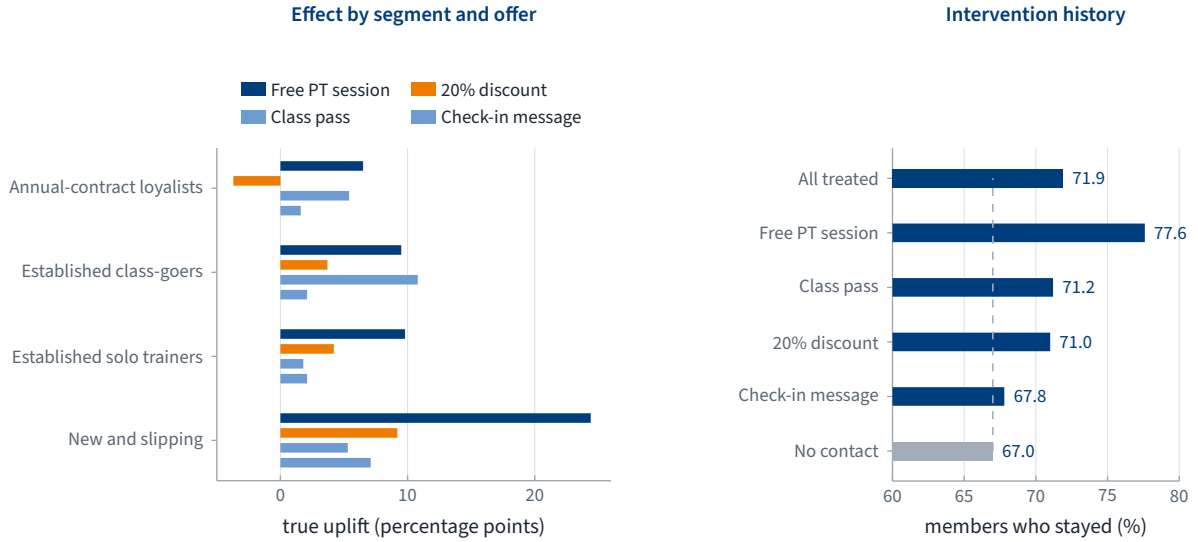


Figure 4 The scenario's ground truth. Left: mean true uplift by segment and offer; a discount backfires on annual members. Right: share who stayed in each arm of the randomised history against the 984 controls (dashed).

Source: `results/a5_true_uplift_by_segment.csv`; `a_scenario_uplift.ipynb`.

member's five drivers average the logistic-regression term with LightGBM's TreeSHAP value [10] (Figure 6). A T-learner [11], one outcome model per offer and one for controls, is the uplift benchmark.

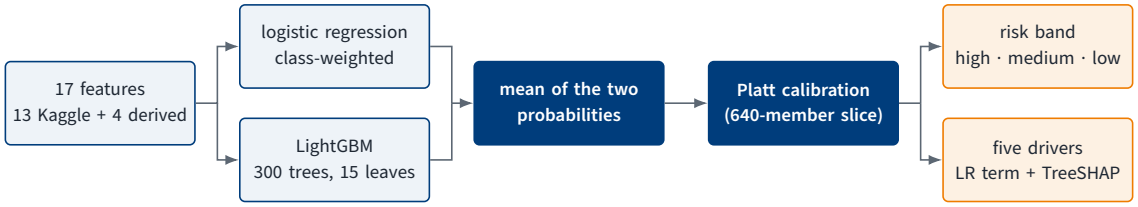


Figure 5 The churn pipeline, fitted on 2,560 training members and calibrated on the remaining 640.

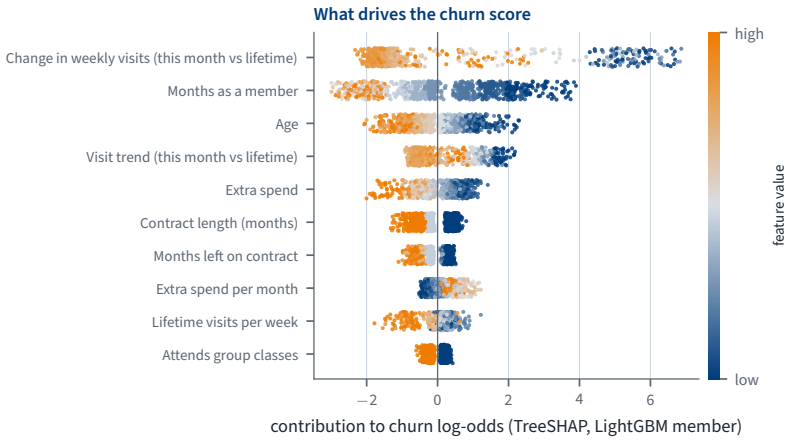


Figure 6 TreeSHAP values of the LightGBM member on the 800 test members, for the ten most influential features. Each dot is a member, coloured by its feature value; dots to the right push towards churn. Real Kaggle data.

Source: `scripts/figures.py` (churn), as in `a_churn.ipynb`, cell 5.

4.2 Decision automation

A forward-chaining engine [12] runs 29 YAML rules: 8 *derive* rules name patterns (falling regular, new member, routine disrupted), 16 *playbooks* propose an offer on a channel, with twins for members without a phone, and 5 *guardrails* block offers (contact and offer caps, eligibility, no paid offers at low risk, no repeat training session). Conflicts resolve by salience, specificity and order; each rule fires once, and blocks are applied after the loop, so guardrails always win. Every rule cites its source, a validator rejects inconsistent rule bases, and administrators edit, test, version and roll back rules in the application. Thresholds were drafted on the training data and adjusted after three *simulated* manager interviews, which are disclosed as plausible opinion, not evidence.

4.3 Decision and optimisation

Each allowed member–offer–channel option is valued as $v_{mo} = \hat{u}_{mo} f_m H - c_o$, the estimated uplift [13] times the monthly fee over a cautious six-month horizon, minus the offer's cost. The uplift comes from case-based reasoning [14] (Figure 7): retrieve the 200 nearest past cases in one shared neighbourhood of treated and untreated members, reuse it as a local A/B test, revise by empirical-Bayes shrinkage towards the offer's average [15], and retain logged outcomes as evidence but never in the averages, because the planner's own choices would bias them. A null test with shuffled offers rejected an earlier design that measured risk instead of effect. The plan maximises total value within budget, staff minutes, trainer minutes and stock, plus two effects between members who train together: *spillover* and *diminishing returns* per instructor. These make the problem non-linear, so a genetic algorithm (DEAP [16]) searches it with a repair operator and a greedy seed, while CP-SAT [17] solves the linear case exactly and benchmarks the full one on small instances. A policy evaluator grades plans against the true effects and is never used to choose one.

4.4 Knowledge graph and staff assistant

A Neo4j knowledge graph [18] links members, classes, instructors, clubs, offers and events through read-only parameterised Cypher. It flags members whose anchor instructor left or whose class was cancelled, as rule facts cached so the rules work without Neo4j; it gives the optimiser its class cohorts; and it simulates an event's impact without changing data (Figure 9). The staff assistant routes a

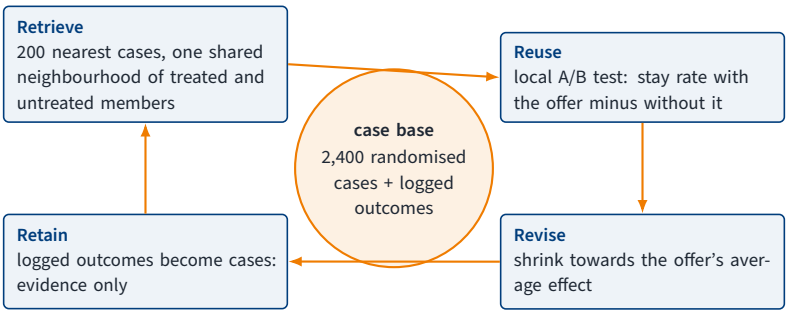


Figure 7 The case-based reasoner. The neighbours that produce an estimate are shown to staff as its precedents.

question to one of seven intents with sentence-embedding centroids [19] and an 'unknown' threshold, fills slots, runs one deterministic tool, and lets the language model only phrase the numbered evidence with citations, or answer that it does not know (Figures 8 and 10). Without the model, it returns the cited evidence itself.

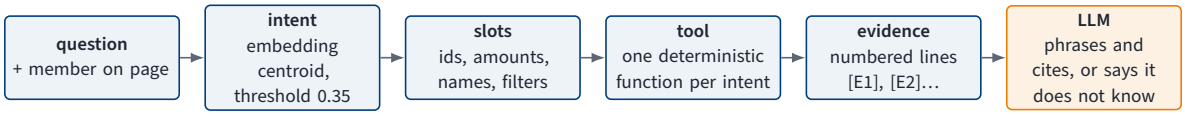


Figure 8 The assistant pipeline: the language model never chooses a tool and never sees data a tool did not return.

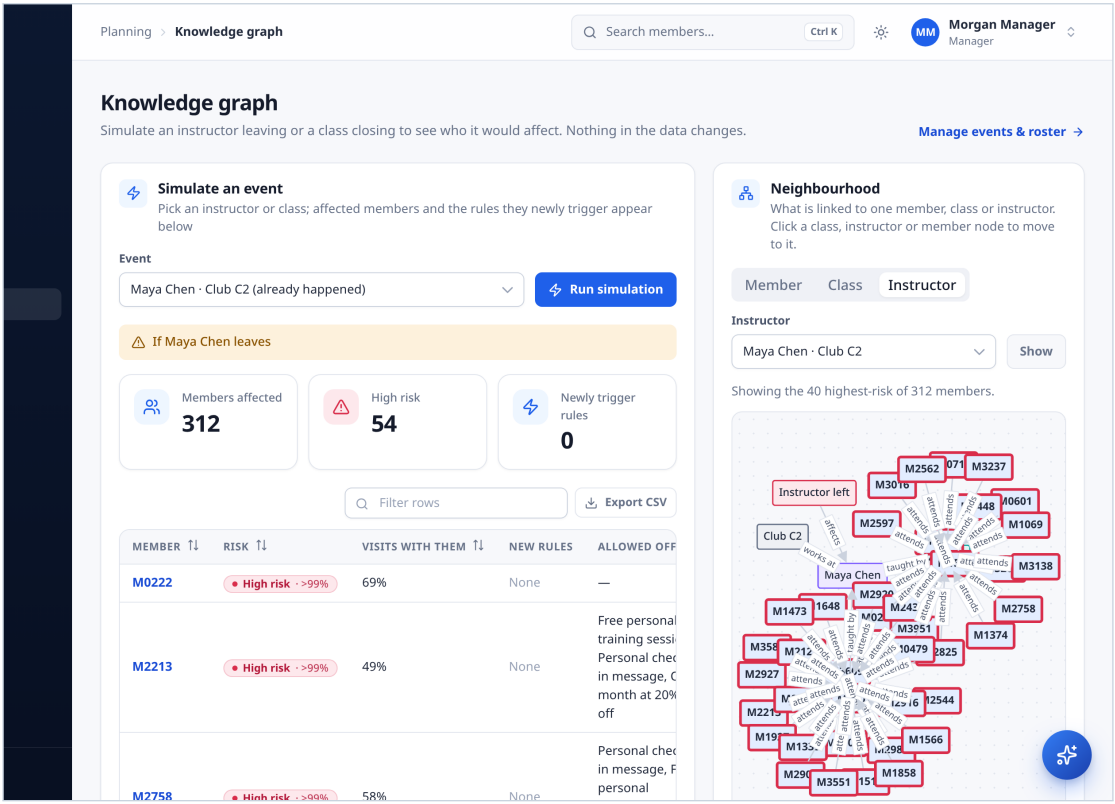


Figure 9 Simulating the departure of instructor Maya Chen in the graph explorer: 312 members are affected, 54 of them at high risk. Right: her neighbourhood, with the 40 highest-risk of her 312 members drawn.

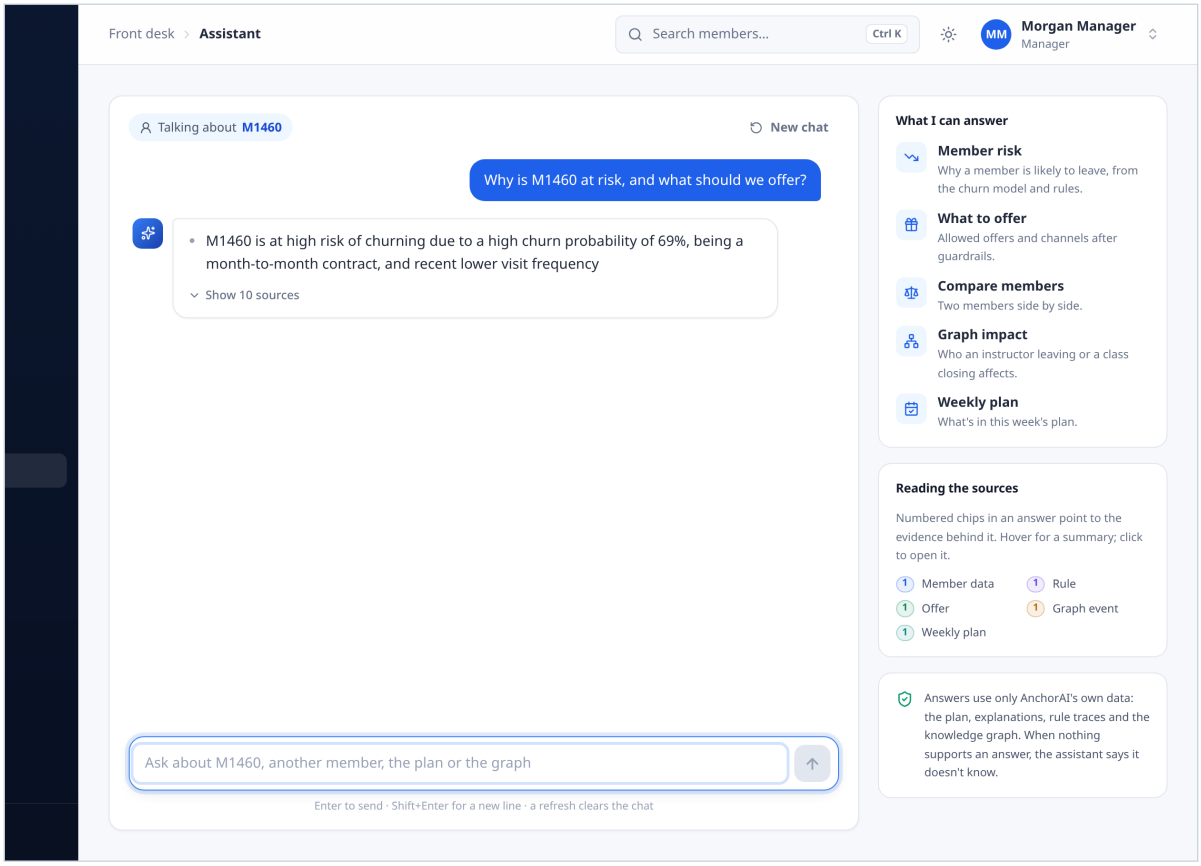
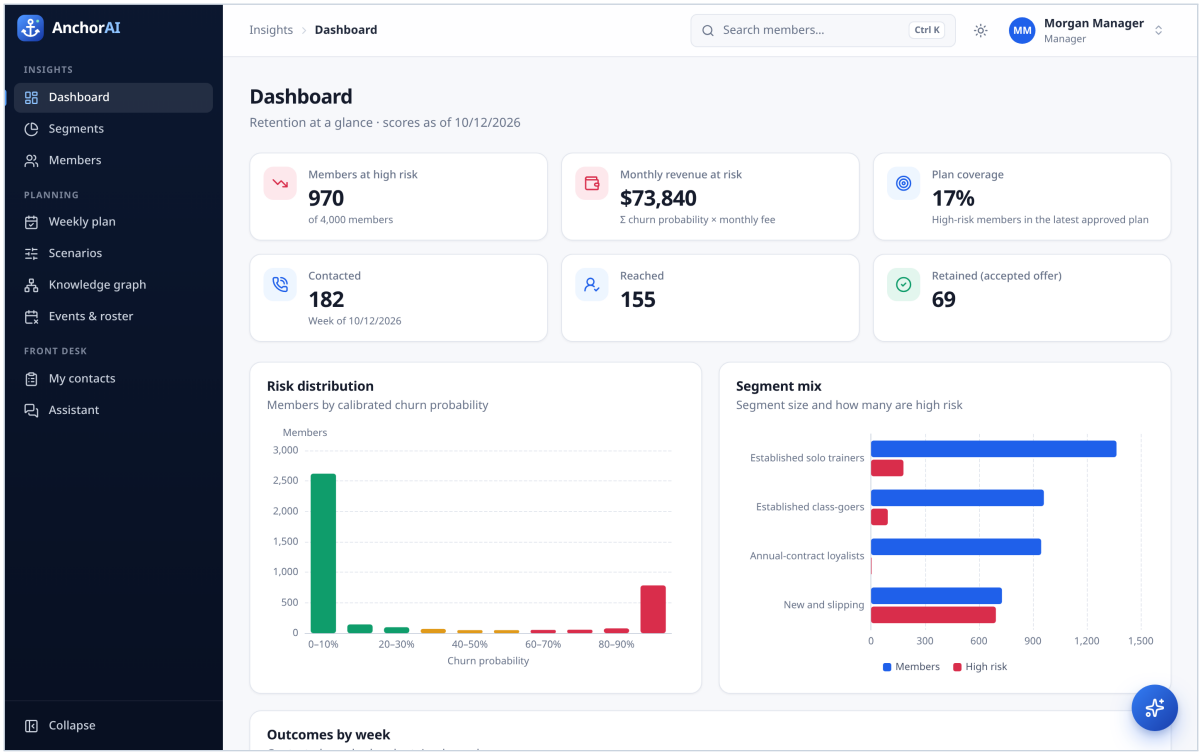
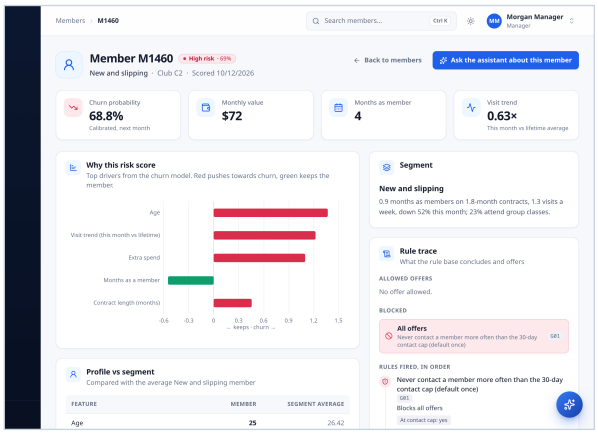


Figure 10 The assistant answering a manager about member M1460 with the local `qwen2.5:7b-instruct` model. The answer cites its sources, which open as evidence chips linking to the member, rule or case.

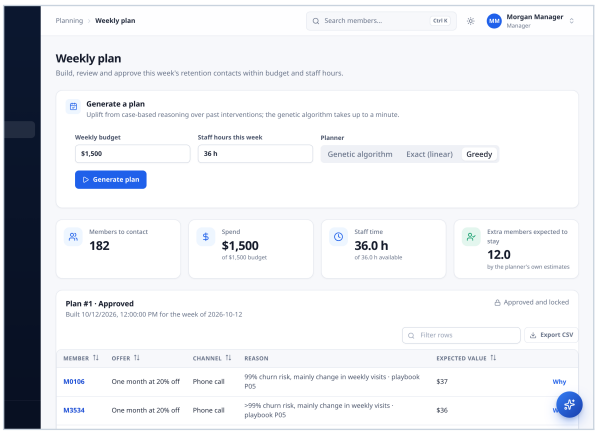
5 System development and implementation

One command builds the data and models, migrates and seeds the database with a demo week, and loads the graph. Sixteen pages serve three roles (Figure 11): managers work from the dashboard, plan and scenarios; staff from a call sheet with talking points, message drafts and two-tap outcome logging; administrators from versioned rules, offers, data and models, users and an audit log of every write. Member 360 shows every component's reasoning about one member on one screen. Risk is always shown as text and colour; planning continues when Neo4j or the language model is down; and 195 offline tests guard against leakage, infeasible plans, rule order, unsafe queries and role or audit errors.





(a) Member 360: risk, drivers, profile and rule trace for one member.



(b) The weekly plan: 182 approved contacts, each with a reason.

Figure 11 The application on the demo data. Top: the manager's dashboard.

6 Experiments and findings

E1 and E2 use only real members; E3–E7 are graded by the scenario's ground truth, which no estimator or planner reads; E8 evaluates the assistant on the system's own evidence (Table 2).

Segments and churn (E1, E2). Silhouette is low and flat on behavioural data, but the four segments are stable across seeds and separate churn by a factor of almost 70 (Figure 12); the 728 'new and slipping' members hold about two-thirds of all churners. The calibrated ensemble beats both baselines on every metric (Figure 13); each risk band churns at about its predicted rate (high band 93.9% against 95.1% predicted), and without the two suspect same-month columns F1 falls to 0.817, the honest lower bound.

Table 2 The eight experiments: real-data experiments use the 4,000 Kaggle members; scenario experiments are graded against the synthetic ground truth.

	Question	Key result	Verdict
E1	meaningful segments? REAL DATA	$k = 4$ by rule; stability ARI 1.00; churn from 94.6% to 1.4%	● Met
E2	churn model beats simple rules? REAL DATA	F1 0.908, PR-AUC 0.973, Brier 0.035 vs 0.862 / 0.953 / 0.056 (logistic) and 0.549 / 0.390 / 0.336 (visit rule)	● Met
E3	who is persuadable? SCENARIO LAYER	CBR ties the T-learner (Spearman 0.39 each), better for PT and discount	● Partly met
E4	more members per dollar? SCENARIO LAYER	22–138% more members kept than by risk or schedule; 0 of 198 plans infeasible	● Met
E5	GA good and fast enough? SCENARIO LAYER	within 0.24% of the optimum on small instances; 33 s for 2,000 members	● Partly met
E6	rules correct; guardrails matter? SCENARIO LAYER	0 validator issues; 29 of 29 cases; guardrails remove 31.3% of proposals	● Met
E7	graph signals add information? SCENARIO LAYER	no ranking gain; 243 extra low-risk members reached; +7 members at 120 h	● Partly met
E8	assistant accurate, grounded? SCENARIO LAYER	96.2% of sentences supported; held-out routing 50.0% vs 54.2% for keywords	● Partly met

Source: `SystemCode/notebooks/results/`; full method and caveats per experiment in the full report.

Recall for the rare churners aged 31 and over is 0.52, which we report with a mitigation rather than hide.

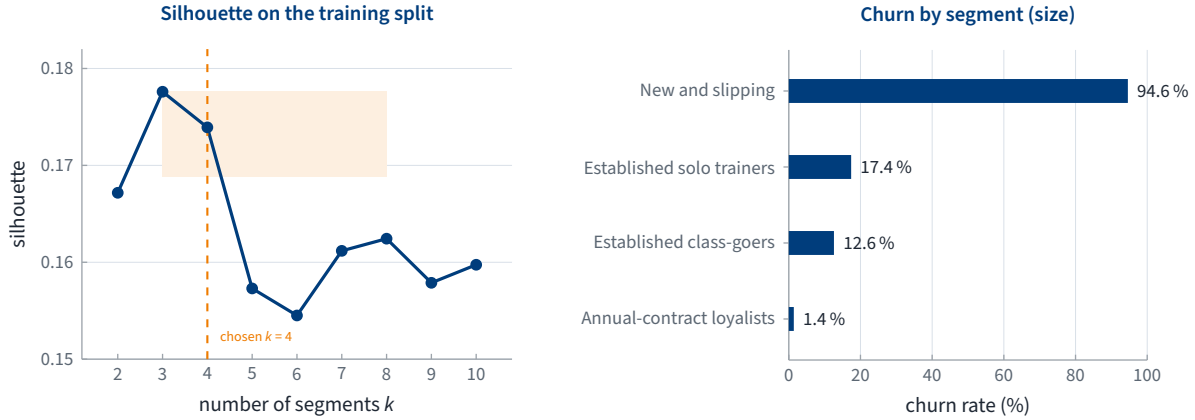


Figure 12 E1. Left: silhouette is low and nearly flat; the shaded band is within 5% of the best k between 3 and 8, and the rule picks the most detailed k inside it. Right: the four segments separate churn from 94.6% to 1.4% (sizes 728, 1,365, 961 and 946).
Source: `results/e1_silhouette_by_k.csv`, `results/e1_segment_profiles.csv`; `a_segmentation.ipynb`.

Uplift and plans (E3–E5). Case-based reasoning matches the T-learner overall and is better where effects are large (Figure 14), but cannot rank members for the small-effect check-in and class pass, which costs 3.7 members a week at the default 36 staff hours (21.0 against 24.6 extra members; level or ahead with more staff time). Planning by effect keeps 1.9 to 3.5 times as many members as contacting by schedule or risk with perfect knowledge (Figure 15) and 22–138% more on estimates. On noisy CBR estimates, however, the genetic algorithm keeps fewer members than greedy: the optimiser's curse [20], where stronger search selects estimates that happen to be too high. On its own objective the algorithm is near-optimal (0.8–2.3% from CP-SAT's optimum on the linear problem; within 0.24% on small full

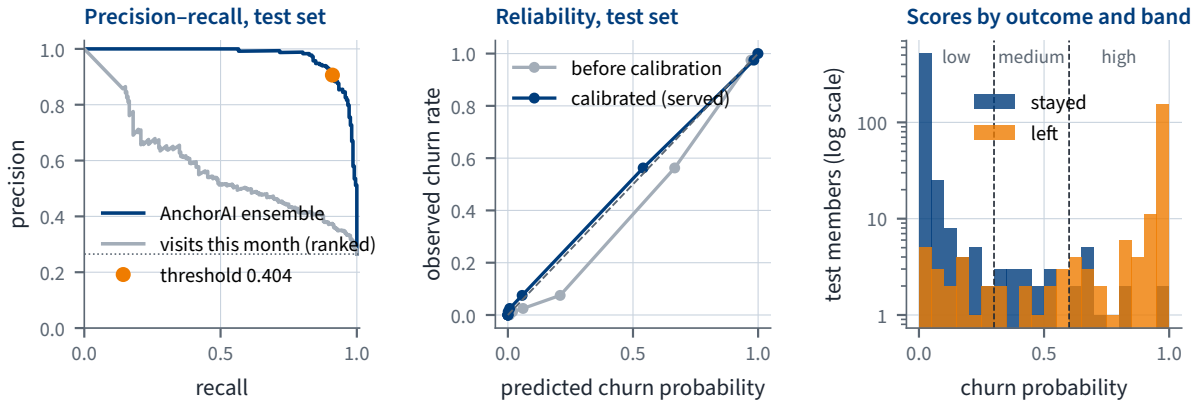


Figure 13 E2 on the 800 test members: precision–recall against ranking by this month's visits, reliability before and after calibration, and scores by outcome with the band cut-offs.

instances) and robust to the assumed spillover and diminishing-returns strengths, but CP-SAT is the better tool for the linear problem.

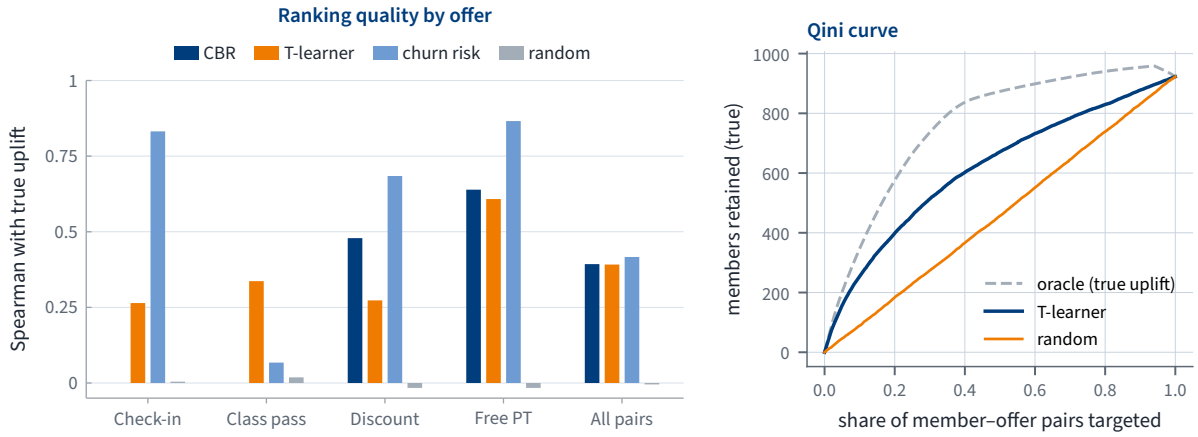


Figure 14 E3. Left: rank correlation between estimated and true uplift; CBR gives every member the offer's average for the check-in and the class pass, so it has no ranking there. Right: Qini curve of the T-learner against the oracle and random targeting.

Source: `results/e3_uplift_vs_truth.csv`, `results/a6_tlearner_recovery.csv`; `b_cbr.ipynb`, `a_scenario_uplift.ipynb`.

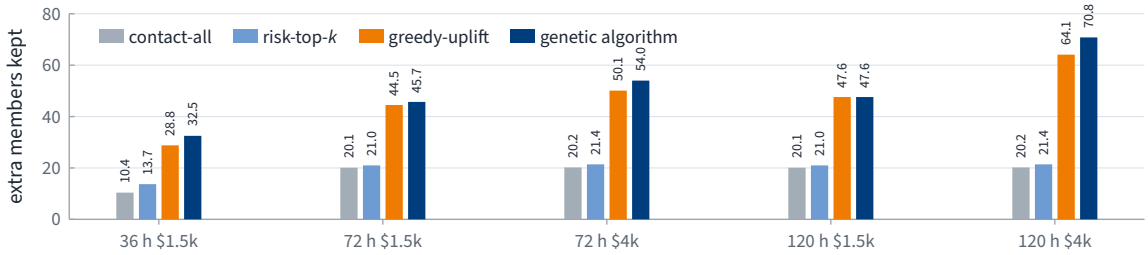
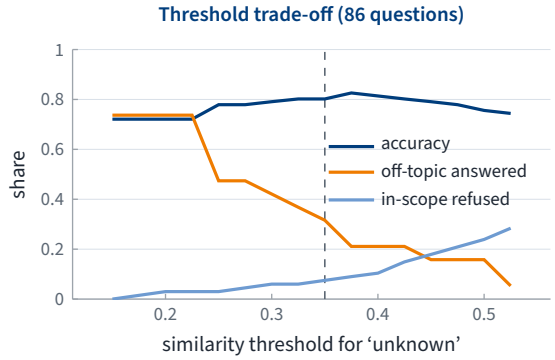


Figure 15 E4 with perfect knowledge: extra members kept by each planner, by staff hours and weekly budget.

Source: `results/e4_optimiser.csv`, mean over three seeds.

Rules, graph and assistant (E6–E8). The rule base is consistent, and its guardrails remove 31.3% of proposals, almost all repeat contacts within 30 days, at little cost where staff time binds; the scenario has no fatigue effect, so we claim safety, not retention. The graph's flags recover the designed event effects (hazard ×1.80 and ×1.32) but do not improve the risk ranking; they route 243 low-risk members

with a disrupted routine to a free check-in, worth about 7 members when staff time allows. The assistant states only what its evidence supports (no invented number; 15 of 17 answers fully supported), but on 24 questions written by team members who never saw its training examples it routes no better than a keyword router (Figure 16), mostly by answering off-topic questions instead of refusing.



Questions	n	Classifier	Keywords
AI-written	62	**91.9**	77.4
Held out	24	50.0	**54.2**

Intent accuracy (%).

Figure 16 E8: raising the threshold refuses more off-topic questions but also more in-scope ones; on held-out questions the classifier is no better than keywords.

Source: `results/e8_threshold_sweep.csv`, `results/e8_intent_predictions.csv`.

7 Findings, discussion and future work

Findings. AnchorAI shows that a gym's weekly retention decision can be automated and explained end to end. Real data supports a calibrated churn model and interpretable segments, with most churn in one recognisable group of new members whose visits are falling. Planning by estimated effect keeps substantially more members than planning by risk, but the quality of the uplift estimates, not the strength of the search, limits the planner. The rules and graph add safety and reach rather than accuracy, and the assistant is faithful to its evidence but routes unseen questions no better than keywords.

Discussion. The limits are as clear as the results. The effects that grade the plans are synthetic and grow with risk by design; the real data is one undated snapshot, so leakage of the same-month columns cannot be settled; the interviews were simulated and no gym staff used the system; and the assistant's held-out set is small. An operator should act on new members whose visits fall, review older medium-risk members by hand, keep visit dates and contact outcomes, and hold back a small random control group each week, preferring the greedy planner until better estimates exist.

Future work. Time-stamped data and randomised contacts to learn real effects; uncertainty-aware planning that stops a strong optimiser amplifying noisy estimates; out-of-scope detection for the assistant; drift monitoring after each rebuild; and a field study with gym staff.

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A Project proposal

The proposal as submitted on 11 September 2026; pages 1–3 of the original, the fourth being blank.

FILING REF:

PROJECT PROPOSAL

Date of proposal: 11/09/2026

Project Title: AnchorAI

Group ID (As Enrolled in Canvas Class Groups): Gate 0

Group Members (name , Student ID):

- ABDULAZIZ ALI S HASANAIN (**A0350856X**)
- ASIRI MOHAMMED YAHYA A (**A0351967N**)
- BINMANEEA KHALID WALEED M (**A0355712E**)

Sponsor/Client: *(Company Name, Address and Contact Name, Email, if any)*

Name: Aramco
Address: Dhahran 31311, Kingdom of Saudi Arabia
Contact Name: Alshahrani, Homoud T
Email: homoud.alshahrani@aramco.com

Background/Aims/Objectives:

Gym members who stop attending cancel weeks or months later. That lag is what we target. The Retention Intelligence System segments members by behavior, predicts who will leave, and decides who to contact given a fixed budget and limited staff hours. Retention is the industry's biggest cost. One in three members leaves yearly, half of new joiners within six months, taking both dues and ancillary spend with them.

Current practice fails three ways:

- Calendar campaigns contact everyone, wasting budget on members who would have stayed.
- Threshold rules ignore trend. Twelve visits a month dropping to four beats a steady two, but rules treat them alike.
- Vendor scores give a number, not a decision. Staff cannot interrogate the flag, so they ignore it.

Our goals:

- Segment members interpretably using unsupervised learning.
- Predict churn with a calibrated gradient boosting and logistic regression ensemble.
- Turn predictions into a weekly action plan where every recommendation carries a challengeable justification.

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Market Analysis:

The vendor landscape splits into two layers. The system of record, covering membership management, booking, payments and access control, is dominated by ABC Fitness including Glofox and Trainerize, alongside Mindbody, Xplor Technologies, Perfect Gym, Virtuagym and Daxko. Above it sits a thinner intelligence layer that bolts on rather than replacing the system of record. Keepme is the clearest example, best known for its predictive Keepme Score, targeting multi site operators and pricing by custom quote rather than published rates. Glofox offers a built in "At Risk" churn report. Current buyer guides for AI in gyms place Keepme at the head of the predictive retention and churn analytics category, with Replify leading AI reception and sales.

Four gaps define our opportunity:

- Competitors score risk but do not publicly expose constrained allocation, meaning the decision of which flagged members to contact and with which offer given a fixed budget and finite staff hours.
- Vendor messaging is uniformly built around targeting high risk members, despite academic evidence that risk ranking is suboptimal.
- Scores trained on tens of millions of profiles are powerful but unauditable, and mid market operators repeatedly cite staff distrust of black box flags.
- The intelligence layer is priced for larger chains. Keepme is generally recommended only once an operator reaches 5+ locations with real lead volume and a sales team to feed, leaving the one to five club operator, the bulk of the market by count, underserved.

We position this project as an explainable, budget aware retention planner for mid market chains, differentiating on transparent decision making rather than on model accuracy, where incumbents with far larger training corpora will always win.

Sources: *Health and Fitness Association 2025 Fitness Industry Benchmarking Report; Keepme and Glofox product documentation; industry buyer guides published in 2026.*
https://www.healthandfitness.org/health-fitness-association-releases-2025-fitness-industry-benchmarking-report/
https://www.keepme.ai/blog/making-the-most-of-your-data
https://www.glofox.com/blog/how-glofoxs-ai-can-mitigate-member-churn-and-boost-retention/
https://www.replify.ai/ai-sales-service-blog/best-ai-for-gyms-complete-buyers-guide

Project Description:

The system is scoped to cover four technique groups. Each group maps to a working component:

- **Decision automation:** A forward chaining rule engine encodes retention playbooks and commercial guardrails staff can read and edit. Fuzzy inference assigns engagement tiers, avoiding threshold cliffs. A case based reasoner picks the offer by matching similar past members and their outcomes.

- **Resource optimization:** Allocating a fixed budget across members, offers and channels is a constrained assignment problem, solved with a genetic

FILING REF:

algorithm. Fitness is expected retained value net of cost, penalized for budget, staff hour, contact fatigue and inventory violations. An exact solver benchmarks small instances so we can report the optimality gap.

- **Knowledge discovery:** K means and Gaussian mixture segmentation, a calibrated churn ensemble, an uplift model for treatment effects, and class recommendations matched to member history.

- **Cognitive techniques:** A knowledge graph of members, clubs, classes, instructors and offers propagates cohort signals such as a cancelled class slot or departed trainer. A natural language assistant answers staff questions, grounded on the rule trace and graph.

Data Collection and Preparation

We will use a hybrid strategy. The public Kaggle dataset of 4,000 members benchmarks our pipeline, but it is a single month snapshot with no intervention data, so trajectory clustering and uplift modeling are impossible on it alone.

We therefore will also build a synthetic longitudinal simulator where each member is an agent with latent habit strength, motivation and social embeddedness, visiting via a process that decays during gaps and strengthens with attendance. It is calibrated to published benchmarks: roughly 66 percent annual retention, a first 90 day cliff, elevated January churn. Treatment effects are deliberately heterogeneous so the uplift model has real structure to recover. All parameters go in the appendix. Rule base and knowledge graph content come from two to four interviews with club managers or trainers, plus published effect sizes.

Dataset: Gym Customers Features and Churn, Adrian Vinueza, CC BY-NC-SA 4.0 https://www.kaggle.com/datasets/adrianvinueza/gym-customers-features-and-churn

Features are built as of a scoring date across recency, frequency, regularity, class engagement, relationship, commercial and lifecycle families. Trend and regularity carry the argument: absolute activity is a weak signal, deviation from a member's own pattern is a strong one.
Two preparation issues:

- **Leakage.** We exclude features generated by the cancellation process itself, such as freeze requests and payment method removal. These are the label in disguise, and likely explain part of the inflated accuracy in existing gym churn papers.
- **Censoring.** Members still active at window end are treated as unobserved, not silently coded as non churners.

Identifiers are pseudonymized, no health status is inferred, fairness is checked across age and gender, and contact fatigue caps are hard constraints in the optimizer.

B Mapping to MR, RS and CGS

All four technique groups the module requires are covered: decision automation, resource optimisation, knowledge discovery and cognitive systems (MR Machine Reasoning, RS Reasoning Systems, CGS Cognitive Systems; D is the course day). The full report also lists every change from the proposal and its reason.

Table 1 Techniques as built, by course and evidence.

Component	Technique as built	Course	Evidence
Data preparation	stratified split; SMOTE-NC inside folds compared with class weights; CRISP-DM	RS D2	E2
Features	trend and ratio features, scaling, leakage check	MR D4, RS D2	E2, leakage check
Segmentation	K-means, silhouette, stability (adjusted Rand index), profiles	MR D4	E1
Churn model	class-weighted logistic regression and LightGBM ensemble, Platt calibration; F1, PR-AUC, Brier	MR D2, D4	E2
Driver explanations	averaged logistic-regression terms and TreeSHAP contributions per member	MR D3, D4	E2 drivers
Scenario layer	synthetic gym and randomised history with known effects	MR D3	scenario layer
Uplift baseline	T-learner (self-researched)	MR D4	E3
Offer reasoner	case-based reasoning: retrieve, reuse, revise (shrinkage), retain	MR D2	E3
Plan optimiser	genetic algorithm with repair, elitism, spillover and diminishing returns	RS D4	E4, E5
Exact benchmark	CP-SAT constraint optimisation	MR D1, RS D1	E5
Policy evaluation	simulation-based comparison against three baselines	RS D1	E4
Rule engine	forward chaining over derive, playbook and guardrail rules; salience; traces; versions	MR D2, RS D1	E6
Knowledge graph	Neo4j and Cypher; multi-hop cohort signals; read-only parameterised queries	MR D3, RS D5	E7
Intent detection	sentence-embedding centroids with an unknown threshold; slot filling	CGS D1	E8
Grounded generation	deterministic tool, then an LLM phrases cited evidence only	CGS D1, D3	E8
Assistant evaluation	keyword baseline; static checks; calibrated LLM-as-judge; local and hosted models	CGS D2, D3	E8
Message drafts	LLM drafts with a rule check and a template fallback	CGS D1	drafts in the application
Overall	hybrid machine learning, rules, optimisation, knowledge graph and language interface with human review	RS D1, CGS D1	E4, E6, E7

C Installation and user guide

Install. Prerequisites: Git, uv 0.13+, Node.js 20+, Docker with Compose, and optionally an OpenAI-compatible LLM endpoint (the default is a local Ollama `qwen2.5:7b-instruct`). Download *Gym Customers Features and Churn* from Kaggle [4] into `SystemCode/data/raw/`; the data is CC BY-NC-SA 4.0 and not redistributed.

```
git clone <repo-url> AnchorAI && cd AnchorAI
cp .env.example .env                    # set ANCHORAI_JWT_SECRET and LLM_*
docker compose up -d neo4j              # knowledge graph
cd SystemCode/backend && uv sync
uv run anchorai setup                   # data, models, seeded DB, graph
uv run uvicorn anchorai.api.main:app  # API on :8000
cd ../frontend && npm install
npm run dev                             # web app on http://localhost:5173
```

Sign in as `admin`, `manager`, `staff1` or `staff2` with the password `anchorai`. The pipeline, API and web application run without Neo4j or a language model, with the graph panels and assistant shown as unavailable. `uv run pytest -q` runs the 195 offline tests; `uv run anchorai db reset` restores the demo database; the administrator's Settings page checks every service.

Manager. The *Dashboard* shows members at high risk, revenue at risk, plan coverage and last week's outcomes (Figure 11). On *Weekly plan*, set the budget and staff hours, choose a planner (genetic algorithm, exact or greedy) and select *Generate plan*. *Why* on a row opens the member's drivers, rule trace, gym events and similar past members. Remove, add or swap contacts with a reason (each is rule-checked at once), *Re-optimise the rest*, then *Approve plan*: contacts are assigned to staff by hours and skills, with printable call sheets. *Scenarios* compares five planners on a what-if budget; *Events & roster* records departures and cancellations; *Knowledge graph* simulates an event without changing data.

Staff. *My contacts* lists this week's assigned members in priority order with the offer, channel, reason and talking points. Log each outcome in two taps (accepted, declined, reached, follow-up, no answer); *Draft message* writes an editable message that is never sent automatically. *Member 360* and the *Assistant* answer questions from AnchorAI's own data, with evidence chips behind every sentence.

Administrator. *Rules* edits, tests on any member, versions and rolls back the rule base; *Offers & constraints* sets the catalogue, staff hours and skills, budget and contact caps; *Data & models* replaces the source file, rebuilds with any seed and shows the model card; *Users*, *Audit log* and *Settings* manage accounts, record every change and check service health.

D Individual project reports

Abdulaziz Ali S Hasanain

Individual reflection

To be written by Abdulaziz Ali S Hasanain.

Peer review

Rate each member's contribution, your own included; contributions add up to 100%.

Member	Contribution (%)	Rating (1–5)	Comments
Abdulaziz Ali S Hasanain			
Asiri Mohammed Yahya A			
Binmaneea Khalid Waleed M			

Asiri Mohammed Yahya A

Individual reflection

To be written by Asiri Mohammed Yahya A.

Peer review

Rate each member's contribution, your own included; contributions add up to 100%.

Member	Contribution (%)	Rating (1–5)	Comments
Abdulaziz Ali S Hasanain			
Asiri Mohammed Yahya A			
Binmaneea Khalid Waleed M			

Binmaneea Khalid Waleed M

Individual reflection

To be written by Binmaneea Khalid Waleed M.

Peer review

Rate each member's contribution, your own included; contributions add up to 100%.

Member	Contribution (%)	Rating (1–5)	Comments
Abdulaziz Ali S Hasanain			
Asiri Mohammed Yahya A			
Binmaneea Khalid Waleed M			